From Frozen Release to 161 Validated PDF Snapshots

What the Frontier production campaign did, what it caught, and what its final pass means

GOTHAM PDF reconstruction notes

June 6, 2026

Status: Production calculation and plotting are complete. The immutable campaign report records 161 of 161 calculations and 161 of 161 plot sets as passing, with no reported errors or missing outputs.

1 What we were trying to accomplish

The practical goal was to replace corrupted archived PDF products with reconstructed products made from the saved full-volume AMR snapshots. The campaign covered five simulations and 161 saved snapshots. For each snapshot, the production system had to calculate 76 PDF products, render 222 plot groups, and leave enough provenance to decide later whether an output should be trusted.

The difficult part was not launching jobs. It was preventing a plausible but incomplete result from looking finished. A reducer can exit successfully after reading a duplicated, stale, or incomplete shard set. A plotter can render a partial directory. A directory can contain most of the expected files and still be scientifically unsafe. The release therefore treated publication as a chain of claims that each had to be checked.

2 The simple picture

Here is the one-sentence version: a frozen executable streamed every native AMR leaf into additive histograms, published a snapshot manifest only after whole-input validation passed, and allowed plotting and final campaign acceptance only from those validated manifests.

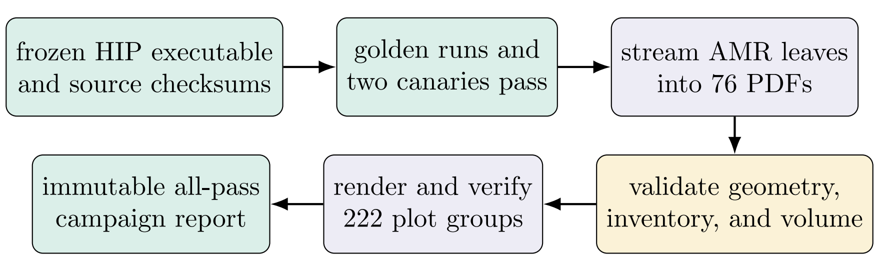

Figure 1: The publication chain. A production result becomes acceptable only after every gate to its left has passed. The diagram is a schematic, not evidence.

The PDFs themselves are additive reductions. For product p, each native AMR leaf cell c contributes a weighted cell volume to one bin:

$$H_p(b) = \sum_c \Delta V_c \, w_p(c) \, \mathbf{1}[B_p(\boldsymbol{q}_c) = b] \, .$$

That structure makes a streaming MPI+Kokkos reducer possible. Each rank reads its assigned shard, bins bounded chunks on the GPU, and participates in global reductions. No global dense simulation cube is assembled.

That only works if the input cells really are the intended snapshot. The production validator therefore checks more than histogram sums. It maps geometry back to logical AMR keys, rejects duplicate leaves and ancestor–descendant overlap, requires the exact contiguous shard inventory, and checks that leaf volumes close to the domain volume. Product payloads are published atomically through temporary `.partial` files, and the snapshot manifest is written last.

3 What was actually done

3.1 A release identity was frozen before production

The production executable and source set were frozen into a release identity. The identity records the HIP executable, source bundle, reducer source, and source-tree revision. Their identifiers begin with `ea53c3cf0f2e`, `0cbb0da68c25`, `ffce7ab54296`, and `db953863b6c0`, respectively; the complete hashes are in the artifact map below.

Production workers did not merely point at an executable path. Before each task, they verified the immutable release identity and five immutable passing reports: three golden runs, a 256-node canary, and a no-control canary. This made "the tested binary" and "the production binary" the same verifiable object.

Gate	Job	Nodes	Shards	Reducer time	Validator
golden original	4767325	128	1024	5.97 s	31 pass, 0 fail
golden science	4767326	128	1024	6.21 s	22 pass, 0 fail
golden all	4767327	128	1024	7.11 s	42 pass, 0 fail
256-node canary	4767328	256	2048	4.39 s	41 pass, 0 fail
no-control canary	4767329	64	512	4.20 s	26 pass, 0 fail

Table 1: The completed Frontier release ladder. Every run also passed geometry-to-logical-key validation and its predeclared performance budget. Source: immutable golden and canary pass reports recorded by the release identity and final traceability record.

3.2 Calculation, plotting, and campaign acceptance were separate gates

For each snapshot, the reducer first verified the input headers and exact shard inventory. During reduction it rejected malformed records, invalid hydrodynamic states, duplicate AMR leaves, and logical-key overlap. It then required domain-volume closure and wrote 76 products. The completed snapshot manifest was the last publication step.

The plot driver accepted only a completed, validated snapshot manifest. It rendered each product's profiles and pairwise heatmaps, wrote per-product metadata, generated an index, and verified the completed plot root. The campaign validator then independently walked the full expected inventory. It checked the calculation identities, source sequences and times, shard counts, geometry-validation flags, product files and finite sums, plot metadata, expected PNG counts, indexes, provenance hashes, audited source views, missing outputs, unexpected outputs, and partial artifacts.

4 The evidence that matters

The final trust anchor is the immutable report `production-campaign.8fd47b0c...49388a.json`. It was generated on June 6, 2026 at 02:32 UTC and has status pass. Its summary is:

Measure	Verified result
Expected and completed calculations	161 / 161
Expected and completed plot sets	161 / 161
Products per snapshot	76
Plot groups per snapshot	222
Total PNG plot groups	35,742
Total native cells processed	11,099,956,051,968
Source payload bytes read	355,202,319,844,512
Reducer elapsed-time range	2.79–71.11 s
Reported errors	0
Missing calculations / plot sets	0 / 0

Table 2: Campaign-level results read directly from the immutable final report.

Simulation	Validated snapshots
`res_4pc`	29
`res_4pc_highmdot`	28
`res_8pc`	49
`res_8pc_lowmdot`	25
`res_8pc_highmdot`	30
Total	161

Table 3: The five-simulation inventory accepted by the final campaign validator.

The report is stronger than a count of Slurm completions. Its campaign validator SHA-256 is itself recorded (`30b4a8137497bb19...d0cf`), as are the release identity, golden and canary reports, plot driver, plotter, snapshot inventory, and the corrected source manifest described below. The report and its SHA-256 sidecar are read-only under the release directory.

5 The failure that mattered

The most useful event in the campaign was a failure. The original source directory for snapshot `res_8pc_highmdot/phase1/00000` contained 250 shard directories. They did not all belong to one snapshot:

- `node_00000000` through `node_00000199` were the contiguous May 12 snapshot;
- `node_00000200` through `node_00000249` were stale May 7 files.

The tempting approach would have been to trust the directory count and process all 250 shards. That run was attempted, and the reducer rejected it because global AMR validation found duplicate logical leaf keys. No manifest was published for the bad input.

The corrected run used an explicit, audited source view containing exactly the 200 current shards. Corrected job 4769652 then validated 25,192 unique AMR leaves, closed the domain volume to 64,000,000.000000089, processed 6,603,931,648 cells, and completed in 4.049 seconds. The final report preserves the reason for the override, the included and excluded node ranges, both mtime ranges, and a SHA-256 over the source-view entry metadata.

Bottom line. This incident is evidence that the validation was active rather than ceremonial. A superficially complete shard directory failed closed; the campaign accepted only the audited full snapshot.

6 What worked, and what did not

What worked:

- the frozen HIP release passed the full 1024-shard golden sequence, 2048-shard scaling canary, and no-control validation before production;
- all 161 intended snapshots produced validated 76-product manifests;
- all 161 snapshots produced verified 222-group plot roots and indexes;
- the final campaign validator found no errors or missing calculations or plots;
- the stale-shard input was rejected before publication and replaced by a traceable audited source view.

Several operational attempts did not work. Frontier submission limits required staged submission. Thirty-two early retries failed immediately because an external helper passed an empty `MANIFEST_PATH`; they created no accepted source or output artifact. One source-view precheck failed because its `find` invocation did not follow directory symlinks; the corrected helper did, without changing the reducer or release checks. These attempts remain visible in the logs, but none were accepted by the final report.

The important engineering result is not that every attempt succeeded. It is that failed attempts could not satisfy the publication contract.

7 What the result means

The completed campaign strongly supports this claim:

> For every snapshot in the declared 161-snapshot production inventory, the frozen reducer processed an exact, globally validated AMR leaf set, published the expected 76 reconstructed PDF products, and the frozen plotting path published the expected 222 verified plot groups. The immutable campaign validator found no missing or inconsistent accepted result.

That is the claim this work was designed to establish. It means the reconstructed products and plots are ready to be used for scientific analysis, with the immutable campaign report as the root of trust.

It does *not* prove every possible scientific interpretation of those PDFs. In particular:

- the same-state bad-line/fixed-line AthenaK A/B run remains the cleanest causal proof of the original archive corruption;
- the independent oracle covers all 14 original products but only 20 of 62 science products;

- the campaign validates this declared reconstruction model and product catalog; it does not validate unrelated magnetic, particle, history, or cooling diagnostics;
- numerical agreement with an online double-precision implementation is not a claim of bitwise identity.

Those are scientific-validation limits, not missing production outputs.

8 How to use the result

Use the immutable campaign report as the first acceptance check, then follow its recorded provenance to the snapshot manifest and per-product plot metadata. Do not infer acceptance from a Slurm exit code or from the presence of a plot directory alone.

The primary artifacts are:

Immutable campaign report
: `/lustre/orion/ast207/proj-shared/gotham/analysis/products/pdf_rebuild/release/`
`production-campaign.8fd47b0c5cc8b405779e37fc865685aa255088bb6aeaa00e389de953ee49388a.json`

Frozen release identity
: `/lustre/orion/ast207/proj-shared/gotham/analysis/products/pdf_rebuild/release/`
`frontier-release-identity.7a517368fe384bc3e793dca9bea96ff0042a5a6995815be1ceb25ab5abd9cf4f.json`

Validated calculation roots
: `/lustre/orion/ast207/proj-shared/gotham/analysis/products/pdf_rebuild/production`

Verified plot roots
: `/lustre/orion/ast207/proj-shared/gotham/analysis/products/pdf_rebuild_plots/production`

Final traceability record
: `/ccs/home/dfielding/athenak-gotham-pdf-rebuild/docs/explanatory_writeups/FINAL_TRACEABILITY.md`

9 Subsequent visualization replacement

The immutable campaign report records the original 35,742-plot release-gated campaign. On June 6, 2026, that visible plot tree was replaced by a user-approved scientific suite: 165 descriptive PNG views for each of the 161 snapshots, or 26,565 PNGs total, plus 1,155 H.264/YUV420p MP4 movies. PNG and movie filenames no longer use opaque `outputXX` labels, and uniform branches are separate movie series.

This replacement did not alter the validated reducer outputs, frozen release identity, or immutable campaign report. The current visualization audits are:

Current plot campaign manifest
: `/lustre/orion/ast207/proj-shared/gotham/analysis/products/pdf_rebuild_plots/production/`
`plot_campaign_manifest.json`

Current movie campaign manifest
: `/lustre/orion/ast207/proj-shared/gotham/analysis/products/pdf_rebuild_plots/production/movies/`
`movie_manifest.json`

10 Bottom line

The full PDF calculation and plotting campaign reached the intended terminal state: 161 validated snapshot calculations, 12,236 PDF products, and 35,742 verified PNG plot groups under one frozen release identity. The strongest reason to trust the result is not the final count. It is that the system rejected a real mixed-snapshot shard directory, required an audited correction, and recorded that exception in the same immutable report that accepts the other 160 snapshots.

The current visible visualization suite is the later descriptive replacement: 26,565 PNGs and 1,155 QuickTime-compatible movies.

The remaining uncertainty is narrower: production completeness is established, but some scientific claims still need broader external-oracle coverage and the clean same-state AthenaK A/B causal test.